⚠️ This website is currently undergoing data validation. Certain datasets are pending review and dependent on updates from the source data team responsible for compiling and delivering the information.

Abraxis Bioscience LLC_10/31/2022

13 minute read

October 31, 2022

Dear Mr. Caforio:

The U.S. Food and Drug Administration inspected your drug manufacturing facility, Abraxis Bioscience, LLC, FEI 3006441852, at 620 N 51st Ave., Phoenix, AZ from March 28 to April 6, 2022.

This warning letter summarizes significant violations of Current Good Manufacturing Practice (CGMP) regulations for finished pharmaceuticals. See Title 21 Code of Federal Regulations (CFR), parts 210 and 211 (21 CFR parts 210 and 211).

Because your methods, facilities, or controls for manufacturing, processing, packing, or holding do not conform to CGMP, your drug product is adulterated within the meaning of section 501(a)(2)(B) of the Federal Food, Drug, and Cosmetic Act (FD&C Act), 21 U.S.C. 351(a)(2)(B).

We reviewed your April 27, 2022, response to our Form FDA 483 in detail and acknowledge receipt of your subsequent correspondence.

During our inspection, our investigators observed specific violations including, but not limited to, the following:

Your firm failed to thoroughly investigate any unexplained discrepancy or failure of a batch or any of its components to meet any of its specifications, whether or not the batch has already been distributed (21 CFR 211.192).

You failed to effectively implement corrective actions and preventive actions (CAPA) and return your aseptic processing line and manufacturing operations to a state of control. Multiple media fill failures occurred between April and October 2021, when simulating aseptic processing operations on the filling line used for your drug product Abraxane (a sterile **(b)(4)** drug).

Your firm failed to perform a sufficiently comprehensive evaluation and remediation upon this loss of state of control in your aseptic operations. In December 2021, you released batches potentially impacted by these failures following **(b)(4)** successful media fill run (conducted in November 2021).

Your investigations into the following media fill failures were inadequate in that they lacked sufficient rigor in determining root causes and scope of impact after the media fills revealed serious non-sterility risks in your aseptic process operations.

Specifically,

1) On April 20, 2021, you conducted a media fill using the **(b)(4)**. On the **(b)(4)** of incubation, you observed a very high number (**(b)(4)**) of contaminated units. Ten different microbes were identified from samples of contaminated units including sporeforming, vegetative, gram-negative, and gram-positive microbes. Your investigation attributed the root cause to **(b)(4)** in the filling manifold that included a damaged **(b)(4)** that was contaminated during **(b)(4)** equipment cleaning step. The investigation indicated that the valve harbored such excessive bioburden that it apparently enabled numerous bacteria to survive the subsequent filling equipment sterilization cycle, and the contaminated valve then caused the extensive contamination of units during the media fill. The investigation failed to adequately address how not only sporeforming microbes, but also vegetative microbes with far **(b)(4)**, presumably survived your validated sterilization cycle. Also, you did not explain whether the sterilization cycle design allows for robust **(b)(4)** and reproducible exposure of all equipment parts in your sterilizer loads.

2) On July 12, 2021, you conducted a repeat media fill using the **(b)(4)**, which revealed multiple **(b)(4)** contaminated units. You attributed the root cause of the media fill failure to personnel gowning, poor disinfection of the inner Restricted Access Barrier System(RABS), and contaminated forceps used during the stopper bowl interventions in your RABS. However, the CAPA did not sufficiently expand to more holistic remediation of the root causes, including but not limited to, ensuring a comprehensive evaluation of the frequency, specificity, sufficiency, and robustness of your disinfection and decontamination program for the RABS and the surrounding cleanroom(s).

3) On October 5, 2021, you conducted a media fill using **(b)(4)** as part of your process improvement. Although you aborted the run due to "unloading equipment failure," you again observed a high number (**(b)(4)**) of contaminated units. Your firm observed this contamination, which was present in **(b)(4)** shelves, while unloading the units from the **(b)(4)**. You attributed the root cause to **(b)(4)** and stated that the bacterial contamination was introduced via the **(b)(4)** performed with **(b)(4)** apparatus that consists of a **(b)(4)** for purposes of sample collection. Notably, your process has incorporated multiple pieces of **(b)(4)** equipment on your sterile processing line rather than traditional reusable

equipment. As with traditional equipment, high standards for robustness and integrity are essential for each of the **(b)(4)** elements of your sterile operation.

The investigation of the **(b)(4)** failure was insufficient. For example, it lacked sufficient evaluation of the ruggedness of these disposable systems, which your firm states are intended to be "**(b)(4)**" while adding that they can be contaminated by the operator manipulations in the ISO 7 (Grade B) environment. While the investigation acknowledges "vulnerability of the syringe sampling system" and inadequate instructions for use of the **(b)(4)** as factors contributing to the contamination of the sterile product process stream, the CAPA did not include a broader review of suitability for their intended use of other **(b)(4)** equipment, adequacy of their instructions, and potential hazards posed when sterile **(b)(4)** are exposed to lower area classifications than ISO 5 (Grade A).

4) On November 15, 2021, you conducted a media fill using the **(b)(4)**, as part of qualification of this new system. You observed one turbid vial on the **(b)(4)** of incubation. You attributed the contamination to inadequate disinfection of the area below the work surface that contaminated the pump apparatus after equipment changeover, which in turn may have contaminated RABS gloves used for unplanned mechanical interventions. However, you did not address the extent of other equipment underneath the filling line worksurface that can pose a contamination risk due to insufficient disinfection.

These **(b)(4)** media fill contaminations indicate your aseptic manufacturing operations were not in control. Without a comprehensive assessment of contamination hazards, and a CAPA that builds more holistic risk mitigation into your operational design, there is no assurance that you can prevent recurrence of sterility problems due to various latent or active failure modes in your operation.

We acknowledge that your firm took significant steps in response to the media fill failures, including a Field Alert Report, batch rejections, and suspension of operations.

Your firm's response is inadequate. You state that "... the media fill investigations determined root cause of each media fill event, supported the establishment of CAPA to prevent future recurrence as well as the decisions made in relation to the product under scope of the events." However, you continued to observe contaminated units in subsequent media fills. The recurring incidents of contaminated units in media fills are an indicator of an adverse trend in your aseptic filling line. You did not provide a comprehensive assessment of contamination risk factors in your aseptic processing operations that support robust and holistic remediations to your aseptic processing operation design. Systemic remediations are essential to address a pattern of recurring contamination events and ensure sustainable control of your aseptic processing operations.

The response also lacks adequate justification to support the sufficiency of your CAPA activities. For example, regarding the October 5, 2021, media fill, your response fails to adequately explain how your CAPA will ensure the **(b)(4)** syringe in your **(b)(4)** will no longer cause contamination of the sterile product pathway. This type of fragility questions the basic functionality and selection of a system that is explicitly designed to maintain sterility. You also did not adequately address whether you

intend to revisit the area classification, the **(b)(4)** equipment choice or supplier of this system considering the media fill finding.

In response to the letter, provide:

- A comprehensive risk assessment of all contamination hazards with respect to your aseptic processes, equipment, and facilities, including an independent assessment that includes, but is not limited to:

  - All human interactions within the ISO 5 area.
  - Equipment placement and ergonomics.
  - Air quality in the ISO 5 area and surrounding room.
  - Facility layout.
  - Personnel Flows and Material Flows (throughout all rooms used to conduct and support sterile operations).
  - Other variables that have potential to cause contamination in your operations (e.g., insufficient **(b)(4)** sterilization frequency).

- A detailed remediation plan with timelines to address the findings of the contamination hazards risk assessment. Describe specific tangible improvements to be made to aseptic processing operation design and control.
- Your CAPA plan to implement routine, vigilant operations and management oversight of facilities and equipment. This plan should include, among other things, improved management of operations that assures prompt detection of equipment/facilities performance issues, timely upgrades to equipment and facilities, adherence to appropriate preventive maintenance schedules, effective execution of repairs, and improved systems for ongoing management review. Your plan should also ensure that appropriate actions are taken throughout the company network.
- A thorough evaluation of the suitability of each **(b)(4)** apparatus for its intended use. Your evaluation of all of your **(b)(4)** equipment should include a determination of whether they are appropriate, durable, and maintain integrity. Also, regarding the media fill failure that involved **(b)(4)** difficulties, include an assessment of sampling robustness based upon its purpose and associated hazards, and discuss the feasibility of a lower risk sampling location.
- A comprehensive, independent assessment of your overall system for investigating deviations, discrepancies, complaints, out-of-specification results, and failures. Provide a detailed action plan to remediate this system. Your action plan should include, but not be limited to, significant improvements in investigation competencies, scope determination, root cause evaluation, CAPA effectiveness, quality unit oversight, and written procedures. Address how your firm will ensure all phases of investigations are appropriately conducted.
- A detailed description of the degree of involvement of quality assurance (QA) staff in evaluation, oversight, performance, and guiding of the content and depth of your investigations. Among

other things, explain whether quality assurance plays a lead role throughout each investigation or only at certain specified intervals. Also explain the role and authority of the QA function in assuring appropriate root cause determinations and identification of an effective CAPA plan.

- A retrospective evaluation by a qualified third party of investigations and failure modes related to the capability of your aseptic processing operation to robustly produce sterile drugs, including, but not be limited to:

o All media fill contamination events, and all sterility positives, since January 1, 2019,regardless of whether the batch was shipped to the United States.
o Identification of all potential failure modes associated with these media fill and sterility positives.
o A detailed evaluation and description of each aseptic connection and operation made down stream of the **(b)(4)**, including, but not limited to, the manipulations at sampling ports in the product flow pathway **(b)(4)**.
o A comparison of your aseptic manufacturing process to the process simulation protocol to identify areas in which media fills may be improved to simulate actual operations more accurately.
o Media fill criteria used by your firm, and adequacy of provisions to ensure thorough investigation of any contamination.
o All changes implemented to your aseptic operations in response to any aseptic process simulation incidents and sterility failures since 2019, including an evaluation of their adequacy and sufficiency, and a risk assessment of any distributed product affected by deficient aseptic processing operations that occurred during this period.
o Your plan to ensure appropriate aseptic practices and cleanroom behavior during production. Include steps to ensure routine and effective supervisory oversight for all production batches. Also, describe the frequency of quality unit oversight (e.g., audit) during aseptic processing and its support operations.

- A thorough retrospective review and risk assessment that evaluates how poor aseptic technique and cleanroom behavior may have affected the quality and sterility of your drugs.
- An independent review of the validity and robustness of your equipment parts sterilizationcycles. This review should include:

o A comprehensive assessment, including load configurations, pre-assembled parts, **(b)(4)**, biological indicator placement in worst-case locations, correlation of heat and biological kill, and other factors.

o A thorough evaluation of the ability of the vegetative microbes, named in your April 2021, media fill investigation to survive the **(b)(4)** sterilization cycle used for equipment parts. The review should also re-evaluate whether vegetative microbes may have been introduced to that media fill through other routes of contamination.

- A complete list of complaints received for Abraxane drug products since January 2019.Include the lot number, date of manufacture, sufficient narrative regarding the nature of the complaint,

any CAPA steps taken, and conclusions.

- An updated risk assessment that addresses any product quality or patient safety risks related to batches of Abraxane (manufactured between October 2020 and April 2021) in U.S. distribution, including potential customer notification and recalls.

Additional Guidance on Aseptic Processing

See FDA's guidance document *Sterile Drug Products Produced by Aseptic Processing—Current Good Manufacturing Practice* to help you meet the CGMP requirements when manufacturing sterile drugs using aseptic processing at .

CGMP Consultant Recommended

Based upon the nature of the violations we identified at your firm, we strongly recommend engaging a consultant qualified as set forth in 21 CFR 211.34 to assist your firm in meeting CGMP requirements.

Your use of a consultant does not relieve your firm's obligation to comply with CGMP. Your firm's executive management remains responsible for resolving all deficiencies and systemic flaws to ensure ongoing CGMP compliance.

Conclusion

The violations cited in this letter are not intended to be an all-inclusive list of violations that exist at your facility. You are responsible for investigating and determining the causes of these violations and for preventing their recurrence or the occurrence of other violations.

If you are considering an action that is likely to lead to a disruption in the supply of drugs produced at your facility, FDA requests that you contact CDER's Drug Shortages Staff immediately, at drugshortages@fda.hhs.gov, so that FDA can work with you on the most effective way to bring your operations into compliance with the law. Contacting the Drug Shortages Staff also allows you to meet any obligations you may have to report discontinuances or interruptions in your drug manufacture under 21 U.S.C. 356C(b). This also allows FDA to consider, as soon as possible, what actions, if any, may be needed to avoid shortages and protect the health of patients who depend on your products.

Correct any violations promptly. Failure to promptly and adequately address this matter may result in regulatory or legal action without further notice including, without limitation, seizure, and injunction. Unresolved violations may also prevent other Federal agencies from awarding contracts.

Failure to address violations may also cause FDA to withhold issuance of Export Certificates. FDA may withhold approval of new applications or supplements listing your firm as a drug manufacturer until any violations are completely addressed and we confirm your compliance with CGMP. We may re-inspect to verify that you have completed corrective actions to address any violations.

This letter notifies you of our findings and provides you an opportunity to address the above deficiencies. After you receive this letter, respond to this office in writing within 15 working days. Specify what you have done to address any violations and to prevent their recurrence. In response to this letter, you may provide additional information for our consideration as we continue to assess your activities and practices. If you cannot complete corrective actions within 15 working days, state your reasons for delay and your schedule for completion.

Please identify your response with the unique identifier: **CMS# 633713**.

Send your electronic response to ORAPHARM4_RESPONSES@fda.hhs.gov with ATTN: CDR Steven E. Porter, Jr. or mail your written response to:

CDR Steven E. Porter, Jr.
Director, Division of Pharmaceutical Quality Operations IV
U.S. Food and Drug Administration
19701 Fairchild Road
Irvine, CA 92612-2506

If you have questions regarding the content of this letter, please contact Jamie Dion, compliance officer, at 303-236-3133, or by email at Jamie.Dion@fda.hhs.gov. www.fda.gov.

Sincerely,
/S/

CDR Steven E. Porter, Jr.
Director, Division of Pharmaceutical Quality Operations IV

CC: Nathan Trembath (via email Nathan.Trembath@bms.com)
Executive Director, Site Head
Abraxis Bioscience LLC
620 N 51st Ave.
Phoenix, AZ 85043